

# Chapter 1, Section 5

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1. Locate the singular points and sketch the following curves in  $\mathbb{A}^2$  (assume  $\text{char } k \neq 2$ ):

- (a)  $x^2 = x^4 + y^4$ ;
- (b)  $xy = x^6 + y^6$ ;
- (c)  $x^3 = y^2 + x^4 + y^4$ ;
- (d)  $x^2y + xy^2 = x^4 + y^4$ .

| $f$                       | $\partial f/\partial x$ | $\partial f/\partial y$ | $\langle \partial f/\partial x, \partial f/\partial y \rangle = \langle 0, 0 \rangle$ | Sing $f$ |
|---------------------------|-------------------------|-------------------------|---------------------------------------------------------------------------------------|----------|
| $x^4 - x^2 + y^4$         | $4x^3 - 2x$             | $4y^3$                  | $(0, 0), (\pm 1/\sqrt{2}, 0)$                                                         | $(0, 0)$ |
| $x^6 + y^6 - xy$          | $6x^5 - y$              | $6y^5 - x$              | $(0, 0), (6^{-1/4}, 6^{-1/4})$                                                        | $(0, 0)$ |
| $x^4 + y^4 - x^3 + y^2$   | $4x^3 - 3x^2$           | $4y^3 - 2y$             | $(0, 0), (3/4, 0),$<br>$(0, \pm 1/\sqrt{2}), (3/4, \pm 1/\sqrt{2})$                   | $(0, 0)$ |
| $x^4 + y^4 - x^2y - xy^2$ | $4x^3 - 2xy - y^2$      | $4y^3 - 2xy - x^2$      | $(0, 0), (3/4, 3/4)$                                                                  | $(0, 0)$ |

(a) is the tacnode, (b) is the node, (c) is the cusp, and (d) is the triple point.

2. Locate the singular points and describe the singularities of the following surfaces in  $\mathbb{A}^3$  (assume  $\text{char } k \neq 2$ ).

- (a)  $xy^2 = z^2$ ;
- (b)  $x^2 + y^2 = z^2$ ;
- (c)  $xy + x^3 + y^3 = 0$ .

| $f$               | $\partial f/\partial x$ | $\partial f/\partial y$ | $\partial f/\partial z$ | Sing $f$             |
|-------------------|-------------------------|-------------------------|-------------------------|----------------------|
| $xy^2 - z^2$      | $y^2$                   | $2xy$                   | $-2z$                   | $(t, 0, 0), t \in k$ |
| $x^2 + y^2 - z^2$ | $2x$                    | $2y$                    | $-2z$                   | $(0, 0, 0)$          |
| $x^3 + y^3 + xy$  | $3x^2 + y$              | $3y^2 + x$              | $0$                     | $(0, 0, t), t \in k$ |

(a) is the pinch point, (b) is the conical double point, and (c) is the double line.

3. *Multiplicities.* Let  $Y \subseteq \mathbb{A}^2$  be a curve defined by the equation  $f(x, y) = 0$ . Let  $P = (a, b)$  be a point of  $\mathbb{A}^2$ . Make a linear change of coordinates so that  $P$  becomes the point  $(0, 0)$ . Then write  $f$  as a sum  $f = f_0 + f_1 + \cdots + f_d$ , where  $f_i$  is a homogenous polynomial of degree  $i$  in  $x$  and  $y$ . Then we define the *multiplicity* of  $P$  on  $Y$ , denoted  $\mu_P(Y)$ , to be the least  $r$  such that  $f_r \neq 0$ . (Note that  $P \in Y \iff \mu_P(Y) > 0$ .) The linear factors of  $f_r$  are called the *tangent directions* at  $P$ .

- (a) Show that  $\mu_P(Y) = 1 \iff P$  is a nonsingular point of  $Y$ .
- (b) Find the multiplicity of each of the singular points in (Ex. 5.1) above.

*Proof.*

- (a) Let  $f'$  be the polynomial  $f(x, y)$  after change of coordinates so that  $P$  becomes the origin and let  $O = (0, 0)$  be the origin. Then,  $P$  is a nonsingular point of  $Y \iff (\partial f/\partial x)(P), (\partial f/\partial y)(P) \neq 0 \iff (\partial f'/\partial x)(O), (\partial f'/\partial y)(O) \neq 0 \iff \partial f'/\partial x, \partial f'/\partial y = \text{higher degree terms} + \text{constant} \iff f'$  has linear terms  $\iff \mu_P(Y) = 1$ .
- (b) (a), (b), and (c) has multiplicity 2, and (d) has multiplicity 3 at  $O$ .

□

## 6. Blowing Up Curve Singularities.

- Let  $Y$  be the cusp or node of (Ex. 5.1). Show that the curve  $\tilde{Y}$  obtained by blowing up  $Y$  at  $O = (0, 0)$  is nonsingular.
- We define a *node* (also called *ordinary double point*) to be a double point (i.e., a point of multiplicity 2) of a plane curve with distinct tangent directions. If  $P$  is a node on a plane curve  $Y$ , show that  $\varphi^{-1}(P)$  consists of two distinct nonsingular points on the blown-up curve  $\tilde{Y}$ . We say that "blowing up  $P$  resolves the singularity at  $P$ ".
- Let  $P \in Y$  be the tacnode of (Ex. 5.1). If  $\varphi : \tilde{Y} \rightarrow Y$  is the blowing-up at  $P$ , show that  $\varphi^{-1}(P)$  is a node. Using (b) we see that the tacnode can be resolved by two successive blowing-ups.
- Let  $Y$  be the plane curve  $y^3 = x^5$ , which has a "high order cusp" at  $O$ . Show that  $O$  is a triple point; that blowing up  $O$  gives rise to a double point (what kind?) and that one further blowing up resolves the singularity.

*Proof.*

- Let  $Y$  be the curve defined by the equation  $xy = x^6 + y^6$ . It is the node of (Ex. 5.1). We consider the equations  $xy = x^6 + y^6$  and  $xu = ty$  in  $\mathbb{A}^2 \times \mathbb{P}^1$ , where  $t, u$  are homogenous coordinates for  $\mathbb{P}^1$ . Now  $\mathbb{P}^1$  is covered by open sets  $t \neq 0$  and  $u \neq 0$ , which we can consider separately. If  $t \neq 0$ , we can set  $t = 1$ , and use  $u$  as an affine parameter. Then we have the equations

$$\begin{aligned} xy &= x^6 + y^6 \\ y &= xu \end{aligned}$$

in  $\mathbb{A}^3$  with coordinates  $x, y, u$ . Substituting, we get  $x^2u = x^6 + x^6u^6$ , which factors. Thus, we obtain two irreducible components, one defined by  $x = 0, y = 0, u$  arbitrary, which is the exceptional curve  $E$ , and the other defined by  $u = x^4 + x^4u^6$ . This is  $\tilde{Y}$ , which meets  $E$  at the point  $u = 0$ . This curve is nonsingular since if we set  $f(x, y, u) = x^4(1 + u^6) - u$ , then

$$\frac{\partial f}{\partial x} = 0, \quad \frac{\partial f}{\partial y} = 0, \quad \frac{\partial f}{\partial u} = -1,$$

so  $\tilde{Y}$  is nonsingular for all points  $t \neq 0$ . The curve  $Y$  is symmetric with respect to  $x = y$ , so  $\tilde{Y}$  is also nonsingular for all points  $u \neq 0$ , hence  $\tilde{Y}$  is nonsingular.

- We can assume  $P$  is the origin so that  $Y$  is defined by  $f = f_2 + f_3 + \cdots + f_r$ . Thus, we can write  $f_2 = (ax + by)(cx + dy)$ , where  $(a, b) \neq (c, d)$  as points in  $\mathbb{P}^1$ . Take  $t, u$  to be homogenous coordinates for  $\mathbb{P}^1$ . If  $t \neq 0$ , substituting  $y = xu$  we have the equation

$$F_u = (a + bu)(c + du) + f'_3 + \cdots + f'_r = 0,$$

where  $f'_i$  is obtained by substituting  $y = xu$  into  $f_i$  and dividing by  $x^2$ , which is possible since  $f_i$  is homogenous of degree  $i > 2$ . This equation defines  $\tilde{Y}$ , which meets the exceptional curve at  $(b, -a)$  and  $(d, -c)$ . Similarly, if  $u \neq 0$ , then we obtain the equation

$$F_t = (at + b)(ct + d) + \hat{f}_3 + \cdots + \hat{f}_r = 0,$$

thus  $(b, -a)$  and  $(d, -c)$  are the only points  $\tilde{Y}$  meets the exceptional curve, that is  $\varphi^{-1}(P) = \{(a, b), (c, d)\}$ . Also,  $\tilde{Y}$  is nonsingular at both points since

$$\frac{\partial F_u}{\partial u} = b\left(c - \frac{ad}{b}\right) \neq 0, \quad \frac{\partial F_t}{\partial u} = b\left(a - \frac{bc}{d}\right) \neq 0.$$

- Let  $Y$  be the plane curve defined by  $x^2 = x^4 + y^4$  and let  $t, u$ , be homogenous coordinates for  $\mathbb{P}^1$ . If  $t \neq 0$ , then we have the equations

$$\begin{aligned} x^2 &= x^4 + y^4 \\ y &= xu \end{aligned}$$

in  $\mathbb{A}^3$  with coordinates  $x, y, u$ . Substituting, we get  $x^2 = x^4 + x^4u^4$ , thus  $\tilde{Y}$  is defined by  $1 = x^2 + x^2u^4$ , which intersects the exceptional curve at  $u = 0$ . Similarly, if  $u \neq 0$ , then  $\tilde{Y}$  is defined by  $t^2 = t^2y^2 + y^2$  and intersects the exceptional curve at  $t = 0$ . These two points correspond to the slopes of the two branches of  $Y$  at  $P$ , which are the two distinct tangent directions on the node  $P$ .

- (d) If  $t \neq 0$ , then we have the equation  $x^2 = u^3$ , which is a cusp. If  $u \neq 0$ , then we have the equation  $1 = t^5 y^2$ , which does not meet the exceptional curve. □

8. Let  $Y \subseteq \mathbb{P}^n$  be a projective variety of dimension  $r$ . Let  $f_1, \dots, f_t \in S = k[x_0, \dots, x_n]$  be homogenous polynomials which generate the ideal of  $Y$ . Let  $P \in Y$  be a point, with homogenous coordinates  $P = (a_0, \dots, a_n)$ . Show that  $P$  is nonsingular on  $Y$  if and only if the rank of the matrix  $\|(\partial f_i / \partial x_j)(a_0, \dots, a_n)\|$  is  $n - r$ .

*Proof.* We follow the hint. The matrix  $\|(\partial f_i / \partial x_j)(a_0, \dots, a_n)\|$  has rank  $n - r$  if and only if there exists a  $(n - r) \times (n - r)$  submatrix with nonzero determinant. The determinant of a matrix of homogenous polynomials where entries of the same row have same degree is also a homogenous polynomial, hence the rank of  $\|(\partial f_i / \partial x_j)(a_0, \dots, a_n)\|$  is independent of the choice of homogenous coordinates of  $P$ . Assuming  $a_0 \neq 0$ , we can pass to an open affine  $U_0 \subseteq \mathbb{P}^n$  containing  $P$  and use the affine Jacobian matrix, where  $U_0$  is the open affine subset consisting of all points with nonzero 0th coordinate. In particular, if we set  $Y_0 = Y \cap U_0$ , then  $g_i(y_1, \dots, y_n) = f_i(1, y_1, \dots, y_n)$  ( $1 \leq i \leq t$ ) generate the ideal of  $Y_0$ . Thus,  $P$  is nonsingular if and only if the rank of the matrix  $\|(\partial g_i / \partial y_j)(a_1/a_0, \dots, a_n/a_0)\|$  is  $n - r$ . Then, we have

$$\frac{\partial g_i}{\partial y_j} = \frac{\partial f_i(1, y_1, \dots, y_n)}{\partial y_j} = \frac{\partial f_i(1, x_1, \dots, x_n)}{\partial x_j}.$$

Since the rank of the Jacobian matrix is independent of the homogenous coordinates chosen for  $P$ , we conclude that  $\|(\partial g_i / \partial y_j)(P)\|$  has an invertible  $(n - r) \times (n - r)$  minor if and only if  $\|(\partial f_i / \partial x_j)(P)\|$  has an invertible  $(n - r) \times (n - r)$  minor. □

10. For a point  $P$  on a variety  $X$ , let  $\mathfrak{m}$  be the maximal ideal of the local ring  $\mathcal{O}_P$ . We define the *Zariski tangent space*  $T_P(X)$  of  $X$  at  $P$  to be the dual  $k$ -vector space of  $\mathfrak{m}/\mathfrak{m}^2$ .

- (a) For any point  $P \in X$ ,  $\dim T_P(X) \geq \dim X$ , with equality if and only if  $P$  is nonsingular.
- (b) For any morphism  $\varphi : X \rightarrow Y$ , there is a natural induced  $k$ -linear map  $T_P(\varphi) : T_P(X) \rightarrow T_{\varphi(P)}(Y)$ .
- (c) If  $\varphi$  is the vertical projection of the parabola  $x = y^2$  onto the  $x$ -axis, show that the induced map  $T_0(\varphi)$  of tangent spaces at the origin is the zero map.

*Proof.*

- (a) Since  $\mathfrak{m}/\mathfrak{m}^2$  is a finite dimension  $k$ -vector space,  $\dim_k T_P(X) = \dim_k \mathfrak{m}/\mathfrak{m}^2 \geq \dim X$  by (5.2A), and by (Ex. 3.12),

$$\dim_k \mathfrak{m}/\mathfrak{m}^2 = \dim \mathcal{O}_{P,X} = \dim X.$$

- (b) A morphism  $\varphi : X \rightarrow Y$  induces a map  $\varphi_P^* : \mathcal{O}_{\varphi(P),Y} \rightarrow \mathcal{O}_{P,X}$  between local rings, where the maximal ideal  $\mathfrak{n}$  of  $\mathcal{O}_{\varphi(P),Y}$  is mapped into the maximal ideal  $\mathfrak{m}$  of  $\mathcal{O}_{P,X}$ . Thus, we can define  $T_P(\varphi)F$  as

$$(T_P(\varphi)F)f = F(d_\varphi f) = F(f \circ \varphi),$$

where  $d_\varphi : \mathfrak{n}/\mathfrak{n}^2 \rightarrow \mathfrak{m}/\mathfrak{m}^2$  is the map induced by  $\varphi_P^*$ .

- (c) By the formula above, it suffices to show  $d_\varphi$  is the zero map at 0. Let  $X$  be the parabola  $x = y^2$  and let  $Y$  be the  $x$ -axis. The maximal ideal  $\mathfrak{m}$  of the local ring  $\mathcal{O}_{0,X}$  is generated by  $y$ , thus  $\mathfrak{m}/\mathfrak{m}^2$  is a one dimensional  $k$ -vector space spanned by  $y$ , and similarly  $\mathfrak{n}/\mathfrak{n}^2$  of  $\mathcal{O}_{0,Y}$  is a one dimensional  $k$ -vector space spanned by  $x$ . Thus, we have  $\varphi^* x = y^2 \in \mathfrak{m}^2$ , hence  $d_\varphi = 0$ , hence  $T_0(\varphi) = 0$ . □